

- 21) Astigmatism
- 22) Repulsion
- 23) Laterally Inverted
- 24) Below the Image of Fish

25) The focal length of the lens = $\frac{1}{1.5} \text{ m}$ (as $P = \frac{1}{f}$ or $f = \frac{1}{P}$)

$$\text{So, } \frac{1}{1.5} \times \frac{10}{10} = \frac{102}{153} = \frac{2}{3} \text{ m} = \underline{0.67 \text{ m}}$$

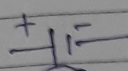
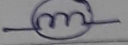
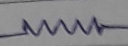
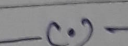
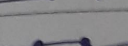
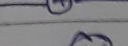
The lens is ^{Converging} ~~diverging~~ or Conver lens.

26) We know $I = \frac{q}{t}$

Given, $I = 5 \text{ A}$, $t = 30 \text{ minutes}$

$$\begin{aligned} q &= I \times t \\ &= 5 \times 30 \times 60 \text{ C} \\ &= 150 \times 60 \text{ C} \\ &= \underline{9000 \text{ C}} \end{aligned}$$

Q7 Circuit symbol of five important components are:-

- 1)  → Cell
- 2)  → Bulb
- 3)  → Resistor
- 4)  → Key
- 5)  → Switch (OFF)
- 6)  → Switch (ON)
- 7)  → Ammeter
- 8)  → Voltmeter

3)

Given: lens nature is convex

Focal length = +12cm

 $h_o = 5\text{cm}$

Object distance = -20cm

Using lens formula,

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\text{or, } \frac{1}{v} - \frac{1}{-20} = \frac{1}{12}$$

$$\text{or, } \frac{1}{v} + \frac{1}{20} = \frac{1}{12}$$

$$\text{or, } \frac{1}{v} = \frac{1}{12} - \frac{1}{20}$$

$$\text{or, } \frac{1}{v} = \frac{5-3}{60} = \frac{2}{60}\text{cm}$$

$$\text{or, } v = \frac{60}{2}\text{cm} \Rightarrow \boxed{30\text{cm}} \rightarrow \text{Image distance}$$

Size of Image $\rightarrow$

$$\frac{v}{u} = m \text{ (magnification)}$$

$$\text{or, } \frac{30}{-20} = m$$

$$\text{or, } \left(\frac{-3}{2} \right)$$

$$\frac{-3}{2} = \frac{h_i}{h_o} \Rightarrow \frac{-3}{2} = \frac{h_i}{5\text{cm}} \Rightarrow +15\text{cm} = 2h_i \Rightarrow h_i = -7.5\text{cm}$$

Hence, the nature of image is real, inverted and magnified.

2) The charge is to be moved from higher potential to lower potential, i.e. $10\text{V} \rightarrow 5\text{V}$.

$$V_B - V_A = \frac{W_{AB}}{q_0}$$

$$\propto, \frac{10\text{V} - 5\text{V}}{\cancel{5\text{V} - 10\text{V}}} = \frac{5\text{J}}{q_0}$$

$$\propto, \frac{5\text{V}}{q_0} = \frac{5\text{J}}{q_0}$$

$$\propto, 5q_0 = 5\text{J}$$

$$\propto, q_0 = \frac{5\text{J}}{5\text{V}} = \underline{\underline{1\text{C}}}$$

PART I

1) Pole: - It is a geometrical centre of the reflecting surface. It is represented by letter P.

Radius of curvature: - The radius of the sphere of which the mirror is a part of.

Principal Axis: - The line joining the Pole and Centre of curvature of the mirror.

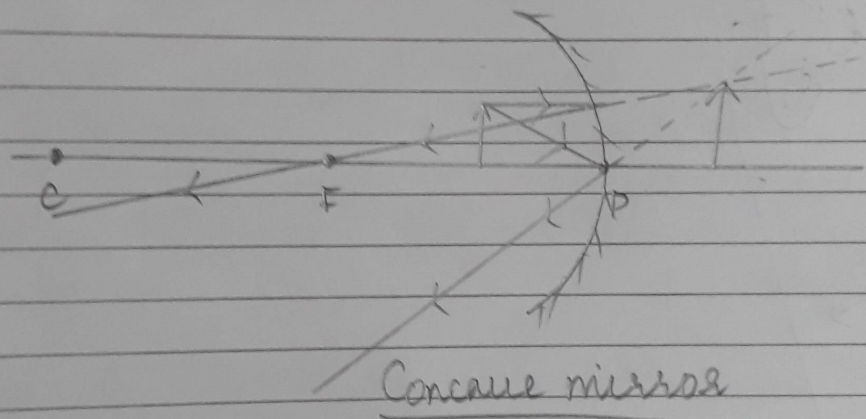
Principal focus: - It is a point on the principal axis where at which rays parallel to principal axis meet after reflection.

PART II → The range of distance is in between the pole and the focus, i.e. 0cm to 20cm.

Characteristics of observed image

- i) Magnified Image
- ii) Formed behind the mirror.
- iii) Virtual Image.

RAY DIAGRAM



- a) The sun appears reddish in the early morning as it is closer to the horizon and undergoes atmospheric refraction.
- b) Power of accommodation of human eye means the ciliary muscle would focus the image on retina when the image is far or near.
- c) The least distance of distinct vision of human eye is 25 cm.